

city needs proactive surveillance, not merely security cameras that let authorities to analyse the scene after a crime takes place. Deep Glint is a solution that uses 3D imaging to monitor large crowds and the way people on the scene behave. When the smart cameras spot a fight, unusual behaviour or doubtful characters, these instantly alert the concerned authorities.

ShotSpotter is another such safety solution. Although gunfire might be something we see only in movies, there are some areas where it is an

Another off-beat initiative in waste management is Zerocycle—a solution that collects and analyses garbage and recycling data to determine recycling rates for all neighbourhoods in a city. This information can be used to prepare customised waste reports for each neighbourhood, which can be circulated among residents. Although it sounds like a simple application, it has been found to be useful in creating awareness about recycling.

Citymapper is a solution that helps travellers to move around comfortably in a new place. It combines

map of emissions at street and building level.

A successful city needs to be inclusive, providing a comfortable environment for those with disabilities. GPS-enabled mobile app BlindSquare is a successful digital service developed from open data in Helsinki. It helps visually-impaired people navigate through the city by describing the environment, announcing points of interest and street intersections, guiding them as they go along.

Awesome Indian products for smart cities

The Indian government's 100 smart cities dream has also spurred a lot of activity in India. The government has made it clear that this dream can be achieved only through public-private synergy. Although there is not much information about what the first 20 chosen cities are doing with their funding, we do hear of random smart developments every day—whether it is a partnership between ISRO and Indian Railways to improve safety, a single dashboard for India's power sector, a system that will speed up tray clearance at the Delhi airport, SMS notifications about delayed trains, increase in digital transactions, tools to speed up Aadhaar verification, or an upcoming drone policy. The start-up economy is also buzzing with new ideas and ventures.

GetMyParking is a Delhi-based start-up that gathers data from parking lots, helping citizens spot and reserve parking. IdeaForge provides solutions for drone-based safety and surveillance, while Flamencotech works on technology for smart buildings and Oizom Instruments develops IoT-based environment monitoring solutions. WeDoSky also works with drones but it customises the intelligence gained from the aerial images for varied purposes like security, irrigation, power lines installation and prediction of an area's solar power potential. Smart Cities Wheel is another Indian start-up that helps urban planners to design more efficient cities using artificial intelligence.

Esysoft Technologies offers software and analytics for smart grids,



Oizom's ambient air quality monitoring system and app help officials and residents stay updated about pollution levels in their area (Source: Oizom)

everyday threat. However, the atmosphere in such areas is shrouded with fear and people hide as soon as gunshot is heard rather than report it immediately. Solutions like ShotSpotter or a network of sound sensors embedded in street lights can be used to report the incident immediately to concerned officials.

Bigbelly offers a smart waste collection and recycling system that has been used in more than 50 countries across the globe. It is not merely an IoT-connected bin but also has a built-in solar-powered waste compacting system that enables the bin to store up to five times the amount of waste as a traditional bin of the same size. When it needs to be emptied, the bin alerts the appropriate city department, helping timely waste clearance and proper optimisation of pick-up truck schedules.

information about the locality's public transport and provides multi-modal transport options to help users get to their chosen destination easily.

Searching for an available parking lot not only kills your precious time but also causes more pollution as you keep circling the neighbourhood! ParkWhiz app helps you find a parking spot in public or private parking lots. It even lets you reserve a paid slot using your credit card.

Cities need actionable information on sources of pollution in order to take corrective and preventive measures. EverImpact is a climate monitoring app that discovers the origins of greenhouse gas emissions in a city. It measures and monetises the city's carbon dioxide emission by mapping ground-level sensor data with satellite locations. City officials get a real-time